

Data Integrity, Sensor Failure Forensics, and Spectral Evidence of the Biosphere 2 LEO Subsurface CO₂ Array

Three Engineered Hillslopes (West, East, Center), 144 Basalt Sensors, 2012–2026

B2 Inverse Dynamics Project
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Abstract

This report audits the integrity of the subsurface carbon-dioxide sensor array installed beneath the three engineered basalt hillslopes of the Biosphere 2 Landscape Evolution Observatory in Tucson, Arizona. Re-deriving every measure directly from the raw records, sampled at a nominal fifteen-minute cadence with documented high-frequency epochs, that span June 2012 to May 2026, we classify all 144 basalt sensors by their most recent behaviour: 46 remain recently alive, 81 are dead, 13 are intermittent, and 4 never produced a usable signal. Signal flicker is near-universal, with 134 of the 144 sensors recording at least one spurious resurrection after falling silent, and no borehole now carries a complete profile of simultaneously live sensors, against a peak of 41 in a single month of 2014. Each channel is classified by its monthly diurnal-amplitude signature, and an independent per-channel spectral verdict, computed for all 144 channels, corroborates that classification on 94 per cent of them, the few disagreements being channels the conservative amplitude gate under-counts; we offer this audit as the evidentiary foundation for any downstream scientific modelling.

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1 Introduction

The Landscape Evolution Observatory at Biosphere 2 in Tucson, Arizona, is among the most ambitious controlled experiments in earth-system science. Inside a climate-controlled glasshouse it houses three nominally identical, fully engineered hillslopes, designated West, East, and Center, each constructed from crushed basalt and instrumented so densely that the movement of water, heat, and carbon through a developing soil can be observed as it happens. The three slopes were built to the same specification precisely so that they could serve as replicates: an intervention applied to one can be read against the others, and a signal common to all three can be distinguished from an artefact of any single one. This replicated, instrumented design is what gives the facility its scientific value, and it is also what makes the health of the instrumentation a precondition for every claim drawn from it.

Among the embedded instruments is a subsurface carbon-dioxide array built from GMM222 sensors seated directly in the basalt. Each slope carries 48 such sensors across 16 boreholes, every borehole instrumented at three depths drawn from four depth horizons (the two shallowest, D1 and D2, present on every borehole, with the deeper D3 or D4 completing each profile), for a total of 144 across the facility. The array samples at a nominal fifteen-minute cadence, with documented departures to one- and five-minute epochs, and has been doing so, in principle, for nearly fourteen years, from June 2012 through the most recent records of May 2026. A continuous, depth-resolved, three-slope record of soil-gas carbon dioxide at this cadence and duration would be a rare and valuable asset, and downstream modelling of gas transport and weathering depends entirely on it being trustworthy.

The purpose of this report is to establish, honestly and from the data themselves, whether that trust is warranted. It is a forensic audit of the array's integrity rather than a scientific interpretation of soil-gas dynamics. We catalogue which of the 144 sensors are still returning usable signal and which are not, we reconstruct the chronology by which sensors failed, and we present spectral evidence that distinguishes a sensor that is genuinely reporting the subsurface from one that is merely emitting numbers. The audit is offered as the necessary first step before any inverse or forward modelling is attempted, because a model fitted to a degraded array will produce confident results that mean nothing.

The audit was motivated by a specific and recurring difficulty. Earlier efforts to carve a clean analysis window out of the record were repeatedly confounded: a sensor that appeared to be reporting would, on closer inspection, be interleaving real measurement with stretches of frozen, repeated, or otherwise phantom values, and a sensor that had fallen silent would briefly flicker back to life only to produce data that could not be physical. This flicker is not a marginal nuisance; it is pervasive across the array, with 134 of the 144 sensors recording at least one such spurious resurrection. Any windowing built on the assumption that a present value is a valid value inherited these artefacts silently. For that reason this report does not trust intermediate or previously cleaned products and re-derives every figure from the raw records at their native cadence. The most consequential single finding follows directly: at the close of the record no borehole carries a complete profile of simultaneously live sensors, where a single month of 2014 saw as many as forty-one, and the decline is documented here sensor by sensor rather than asserted.

2 Data Foundation and Cleaning Protocol

The forensic record rests on a single coherent archive: the complete operating history of the basalt carbon-dioxide monitoring network, persisted as columnar parquet and read as one time series per sensor. The instrumented population comprises 144 basalt sensors, organised as three slopes of 48 sensors each, and the archive spans roughly 13.8 years, from the first sample on 2012-06-28 at 16:30 through 2026-05. Each sensor resolves to a flat schema of timestamp and instrument reading (`datavalue`, in ppm), which is the atomic unit the entire analysis consumes. Because the schema is uniform across slopes, depths and instrument generations, the cleaning protocol described below can

be applied identically to all 144 records, and any departure from the protocol is itself an observable property of the data rather than an artefact of inconsistent ingestion.

The nominal acquisition cadence is one sample every 15 minutes, and for most of the record the data honour that nominal rate. The cadence is not, however, stationary. Measured across six representative sensors and aggregated over 168 months, the archive resolves into five distinct sampling epochs (Figure 1). A brief 5-minute deployment phase (2012-06 to 2012-07) gives way to the steady 15-minute baseline that holds from 2012-08 through 2016-06; a high-frequency 1-minute burst then runs from 2016-07 to 2017-04, followed by a 5-minute interval through the remainder of 2017, after which the network returns to the 15-minute cadence from 2018-01 to the end of the record. Three of these epochs are high-frequency relative to the baseline, and every downstream spectral or diffusion estimate must therefore carry the correct local sampling interval rather than assume the nominal 15 minutes, on pain of mislocating every diurnal harmonic.

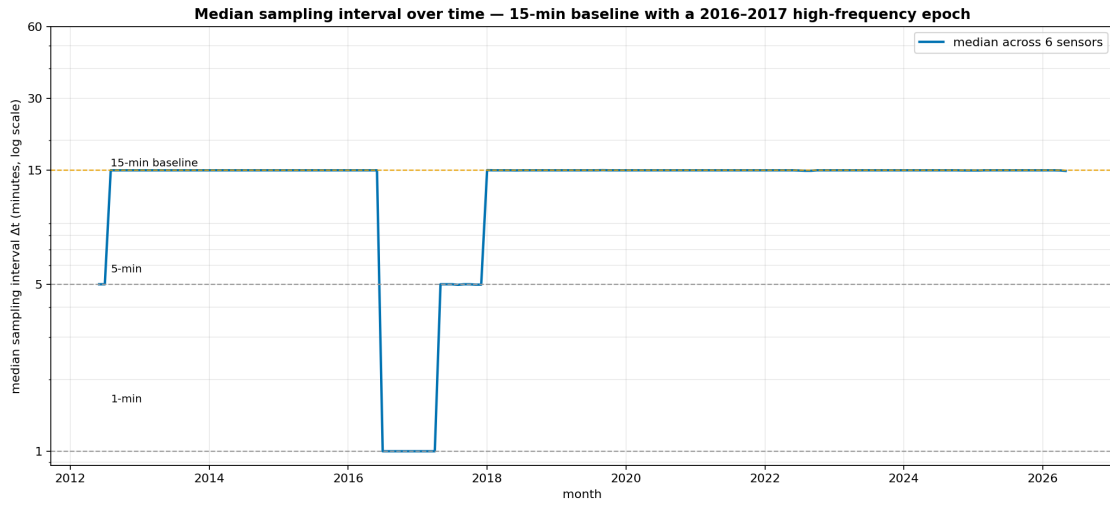


Figure 1: Median inter-sample interval across the archive, showing the five non-stationary sampling epochs around the 15-minute nominal baseline.

Three rules govern what counts as a valid measurement. First, the instrument emits a hard sentinel of -10000 on internal fault; any reading at or below -9999 is the GMM222 error code and is excised, never interpolated as signal. This code is not rare housekeeping but a structural feature: every one of the 144 sensors carries at least one error code, with a median fault fraction of 0.00028, 39 sensors above 0.1%, 4 above 1%, and a maximum of 0.02208 on sensor E_R10_C4_D1. Second, a sustained reading above 3000 ppm is treated as plateau saturation, that is, as a failure mode of the cell rather than a physical concentration; the plausible band for genuine signal is $0 < v \leq 3000$ ppm. The distribution of time spent in saturation (Figure 2) is bimodal: 76 of 144 sensors spend more than 10% of their lives plateaued and several sit essentially permanently above the band, while 32 sensors saturate for under 1% of their record, with a population median plateau fraction of 0.106 and a maximum of 0.999. Third, isolated single-sample excursions surviving the first two rules are removed by despiking, so that the cleaned series retains only physically admissible, temporally coherent values.

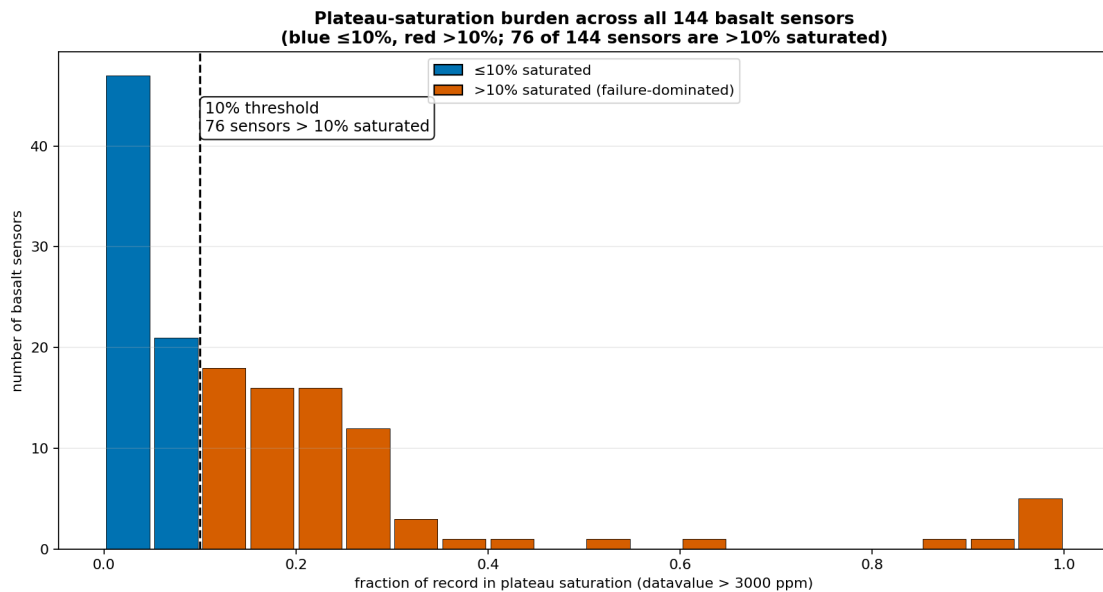


Figure 2: Fraction of each sensor’s lifetime spent above the 3000 ppm saturation threshold, across all 144 sensors. The mass above 0.10 marks sensors for which saturation is the dominant state.

2.1 Why naive windowing fails: a phantom taxonomy

The cleaning rules are necessary because the raw archive contains records that look alive but are not, and a window chosen blindly will sample failure as though it were physics. The behaviour sorts into four archetypes (Figure 3). The first is genuine signal: sensor E_R4_C1_D1 over mid-2015 delivers 4225 clean samples with a mean of 324.9 ppm and a coherent diurnal amplitude of 33.8 ppm, with no error codes and no saturation. The second is sentinel-peppered signal: over a month in early 2013, E_R10_C4_D1 carries 75 isolated -10000 codes embedded in 2996 otherwise plausible samples whose real-signal mean is 192.3 ppm. The third is plateau collapse: W_R18_C-4_D4 sits pinned near 6975 ppm for the entire window, 100% above the saturation band and carrying no error codes at all, so a sentinel filter alone would pass it through as data. The fourth, and most dangerous, is phantom resurrection: C_R4_C4_D2 resurrects 19 times over its life, its dead months marked not by missing data but by collapse of diurnal amplitude while values remain inside the plausible band.

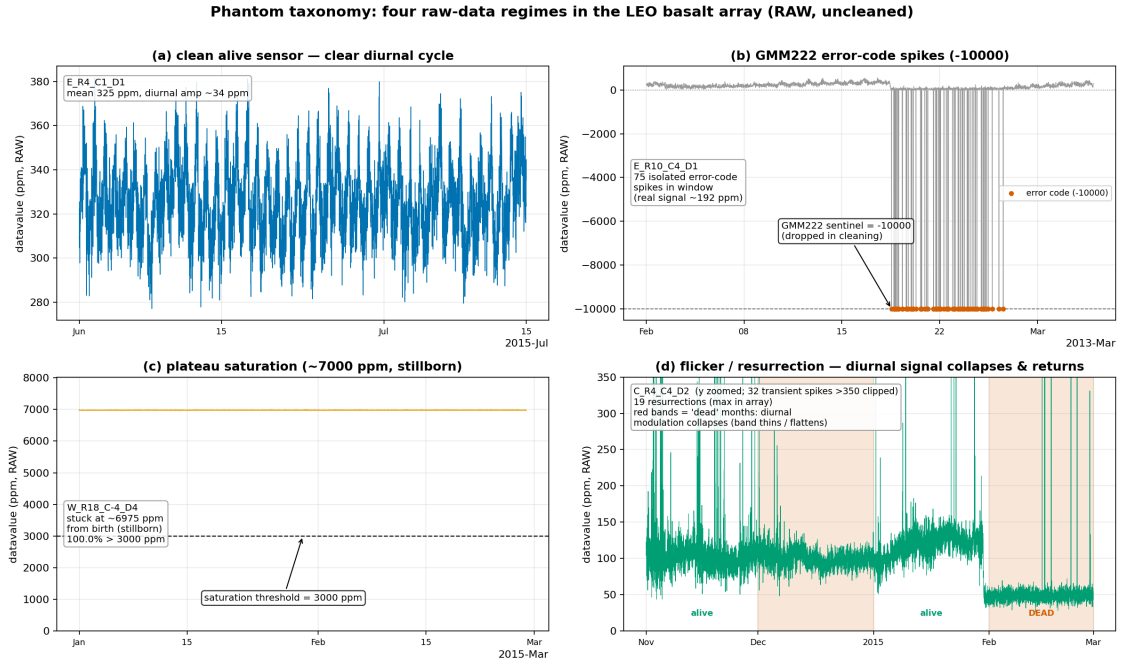


Figure 3: The four data archetypes: clean breathing signal, sentinel-peppered signal, plateau collapse, and phantom resurrection. Only the first is admissible as physics.

This taxonomy is not anecdotal. Across the network, 134 of 144 sensors, or 93.1%, resurrect at least once, the most prolific being C_R4_C4_D2 with 19 resurrections, and 76 sensors spend a substantial fraction of their lives saturated. A window placed without regard to this structure will, with high probability, capture a plateau, a resurrection, or a stretch of sentinel codes and report it as a breathing slope. The phantom taxonomy is therefore the operative reason the cleaning protocol must precede any windowing, and the reason that the forensic analysis in the following sections is built on liveness classification rather than on raw record length.

3 Sensor Liveness and Failure Forensics

A buried CO₂ electrode is never simply on or off; it degrades, saturates, drops out, and occasionally revives, so any honest accounting of the array must first fix what “alive” means and measure it the same way for every channel. We adopt a calendar-anchored monthly liveness rule. A sensor-month is counted as alive if and only if it contains at least ten valid daily records, its median CO₂ concentration falls within the physically plausible band of 100 to 2000 ppm, and its median daily diurnal amplitude reaches at least 8 ppm. The amplitude criterion is the decisive one. A working electrode in a respiring soil column must breathe with the day–night forcing, so a flat trace sitting inside the plausible band is reporting electronics, not soil. The concentration band rejects both disconnected channels reading near zero and saturated channels pinned above the instrument ceiling, while the ten-day floor prevents a single spurious reading from certifying an otherwise silent month.

Recency is judged against the dataset calendar end, 2026-05, and not against each sensor’s own final sample. The distinction is not pedantic. A channel that breathed normally through 2018 and then fell silent has a record that looks complete on its own terms, and a span-relative rule would obligingly call its last recorded months alive. Anchored instead to the common calendar, that same channel is read correctly as dead for the years it has been silent. The array spans 2012-06-28 to 2026-05, roughly 13.8 years, and the only question the commission actually asked is whether each instrument is returning physical signal now; that question has a defensible answer only when the endpoint is fixed for the whole array.

Table 1: Sensor status by slope, anchored to the dataset calendar end (2026-05). A sensor is *alive-recent* if it shows a physical diurnal cycle within the last three months of record, *dead* if its last living month precedes the record end by more than a year, *intermittent* otherwise, and *stillborn* if it never produced a plausible signal.

Slope	Alive-recent	Intermittent	Dead	Stillborn	Total
West	12	3	30	3	48
East	18	6	24	0	48
Center	16	4	27	1	48
All	46	13	81	4	144

Measured this way, of the 144 basalt sensors only 46 are alive-recent, 13 are intermittent, 81 are dead, and 4 never produced a plausible signal and are classed as stillborn. Table 1 resolves the verdict by slope. West has lost the most ground, with 30 of its 48 channels dead and only 12 alive-recent; East retains the largest living cohort at 18 alive-recent against 24 dead; Center sits between them at 16 alive and 27 dead. Figure 4 maps the same classification onto the physical geometry of the three engineered hillslopes, so that surviving and failed channels can be read against their row, column, and depth.

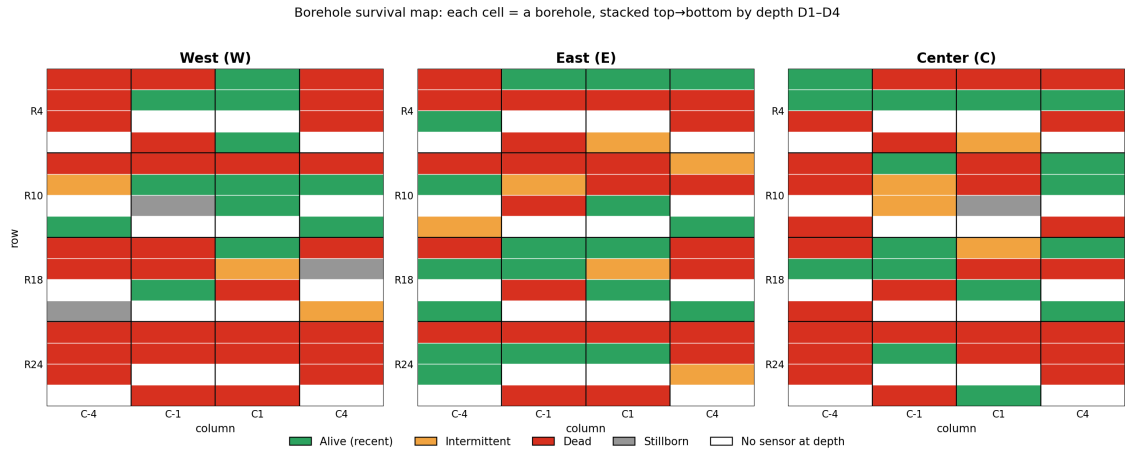


Figure 4: Spatial map of final sensor status across the West, East, and Center slopes. Each cell is one basalt CO₂ electrode at its row, column, and depth position; color encodes the calendar-anchored status (alive-recent, intermittent, dead, stillborn).

The collapse is not uniform in time. The first confirmed death is W_R4_C-1_D4, last alive in 2015-09. West’s mortality opens in 2015-09, East’s in 2016-11, and Center’s only in 2018-07, so the slopes do not begin to fail together. The bulk of the loss arrives much later: the median death month is 2021-08 on West, 2022-03 on East, and 2022-04 on Center, and the single deadliest months cluster between 2020-08 and 2021-08. The most recent death on record is C_R24_C-1_D4 in 2025-05. The cumulative failure curve in Figure 5 therefore shows a long, sparse early tail of isolated 2015–2018 failures followed by a steep mass-extinction wave concentrated in 2021–2022, after which only a thin cohort of survivors remains.

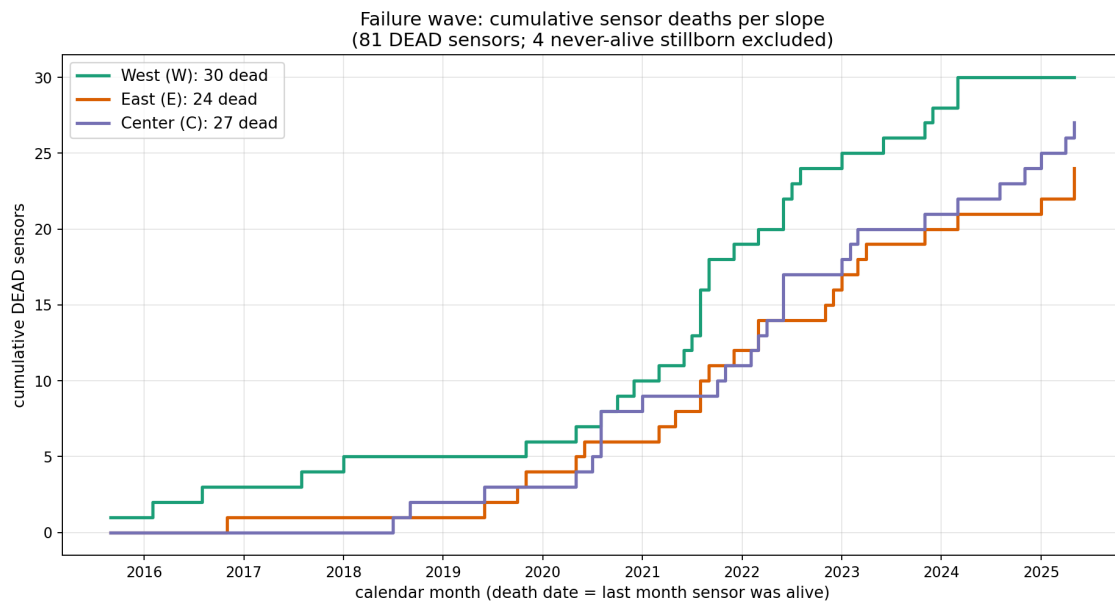


Figure 5: Cumulative count of dead sensors by last living month, anchored to the 2026-05 calendar end. Isolated early failures from 2015 onward give way to a mass wave in 2021–2022; the median death month falls in 2021-08 (West) through 2022-04 (Center).

The depth pattern runs counter to a naive immersion prior and must be reported exactly as the data show it. One might expect the deepest electrodes, furthest from servicing and most continuously immersed, to fail first; they do not. Mortality is highest at the shallowest level: of the 48 depth-1 channels, 70.8% are dead and only 25% remain alive-recent. The deeper levels fare better on the whole, though not strictly monotonically, with depth-2 at 47.9% dead against 39.6% alive, depth-3 (24 channels) at 54.2% dead, and depth-4 (24 channels) the least afflicted at 45.8% dead and 33.3% alive. The defensible statement is therefore that the shallowest horizon is the most-affected on the array while the deeper horizons fare better, with depth-3 exceeding depth-2 so that the decline is not uniform, as Figure 6 makes plain. Far from being inexplicable, a shallowest-worst pattern is precisely what a surface-concentrated moisture mechanism would produce, a reading developed in the cross-slope and concluding sections. A related structural loss compounds the picture: at its 2014 peak a single month saw as many as 41 of the 48 boreholes carry a complete living depth profile, whereas at the close of the record none retain simultaneously live sensors at all three depths, only one borehole (W_R4_C1) still qualifying under the looser alive-recent criterion, so no location retains the full vertical column the array was built to measure.

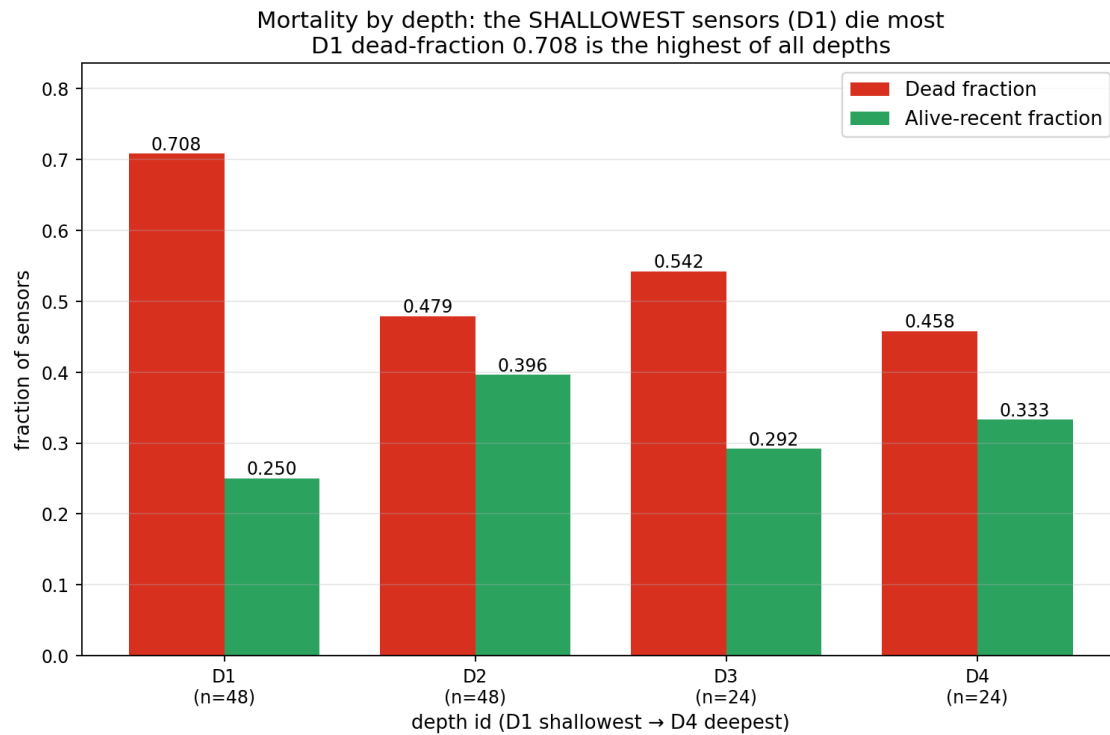


Figure 6: Dead and alive-recent fractions by electrode depth. Mortality is highest at the shallowest level (depth-1: 70.8% dead) and lowest at the deepest (depth-4: 45.8% dead), the opposite of an immersion-driven expectation.

Finally, the array does not fail cleanly; it flickers. Across the 144 sensors, 134 (93.1%) record at least one dead-to-alive resurrection, with a mean of 3.2 and a median of 3 revivals per channel and a maximum of 19 on C_R4_C4_D2. Only 10 channels never resurrect, and four of those are the stillborn units that were never alive to begin with. The distribution in Figure 7 is broad and right-skewed rather than concentrated at zero, which is the signature of a fleet that intermittently saturates and recovers rather than one that simply switches off. Consistent with that reading, 76 sensors spend more than 10% of their raw record pinned in saturation above the plausible band, and every sensor in the array carries logged error codes. This pervasive intermittency is itself diagnostic: it points to recoverable electronic and saturation faults overlaid on the slower true mortality, and it is why the calendar-anchored rule above is necessary to separate a channel that is genuinely returning soil signal from one that merely blinked alive for a single month.

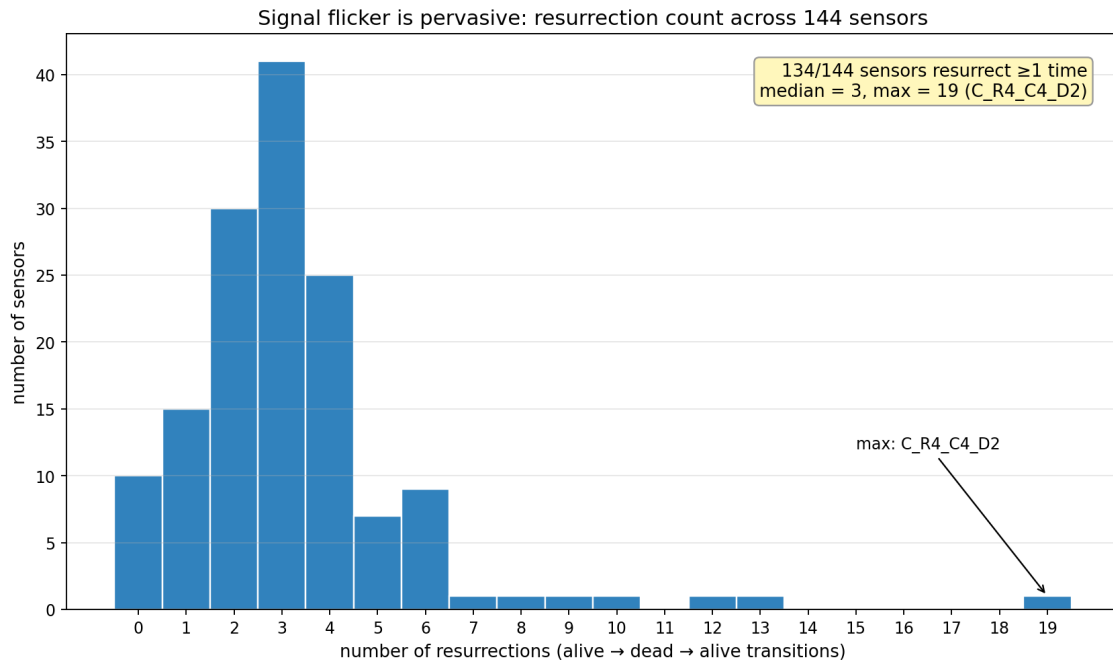


Figure 7: Distribution of dead-to-alive resurrection counts per sensor. 134 of 144 channels revive at least once (mean 3.2, median 3, maximum 19 on C_R4_C4_D2); only the four stillborn units and six others never resurrect.

The complete per-sensor evidence underlying these aggregates is given in Table 2, which lists every one of the 144 basalt channels with its status, depth, first and last living month, longest contiguous alive run, resurrection count, and saturation-plateau fraction.

Table 2: Complete inventory and failure chronology of all 144 basalt CO₂ sensors across the three LEO slopes, sorted by status then last living month. *First/Last* give the first and last calendar month with a physical diurnal signal; *Run* is the longest contiguous alive stretch; *Res.* counts dead→alive revivals (signal intermittency); *Plat.* is the fraction of the raw record spent in >3000 ppm saturation.

Sensor ID	Status	Depth	First	Last	Run	Res.	Plat. %
W_R4_C-1_D4	DEAD	4	2014-09	2015-09	4	3	98
W_R24_C1_D1	DEAD	1	2013-09	2016-02	21	2	91
W_R18_C-4_D1	DEAD	1	2013-09	2016-08	21	3	87
E_R4_C4_D2	DEAD	2	2012-07	2016-11	53	0	63
W_R24_C-1_D1	DEAD	1	2013-09	2017-08	21	3	41
W_R24_C-4_D3	DEAD	3	2013-09	2018-01	19	4	36
C_R24_C-4_D1	DEAD	1	2013-09	2018-07	30	3	34
C_R24_C4_D3	DEAD	3	2013-09	2018-09	30	2	33
C_R24_C-4_D2	DEAD	2	2013-09	2019-06	17	9	30
E_R24_C4_D2	DEAD	2	2012-07	2019-06	57	2	30
E_R10_C-1_D3	DEAD	3	2012-07	2019-10	88	0	26
E_R4_C4_D3	DEAD	3	2012-07	2019-11	89	0	26
W_R18_C4_D1	DEAD	1	2013-09	2019-11	41	3	28
C_R24_C4_D1	DEAD	1	2013-09	2020-05	44	2	26
E_R18_C-4_D1	DEAD	1	2012-07	2020-05	52	2	18
W_R18_C1_D3	DEAD	3	2013-09	2020-05	47	3	26
E_R4_C-4_D2	DEAD	2	2012-07	2020-06	52	1	24
C_R24_C-4_D3	DEAD	3	2013-09	2020-07	43	3	26
C_R24_C1_D1	DEAD	1	2013-09	2020-08	44	3	25
C_R4_C-1_D1	DEAD	1	2013-09	2020-08	50	1	24
C_R4_C-1_D4	DEAD	4	2013-09	2020-08	17	6	26

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Table 2 – continued

Sensor ID	Status	Depth	First	Last	Run	Res.	Plat. %
W_R4_C4_D3	DEAD	3	2013-09	2020-08	50	2	25
W_R18_C-1_D1	DEAD	1	2014-07	2020-10	39	4	53
W_R4_C-4_D2	DEAD	2	2013-09	2020-12	50	3	23
C_R18_C-4_D1	DEAD	1	2013-09	2021-01	30	5	28
E_R4_C1_D2	DEAD	2	2012-07	2021-03	105	0	21
W_R10_C-1_D1	DEAD	1	2013-09	2021-03	45	5	23
E_R18_C4_D1	DEAD	1	2012-07	2021-05	67	1	15
W_R24_C-4_D1	DEAD	1	2013-09	2021-06	60	3	21
W_R10_C1_D1	DEAD	1	2013-09	2021-07	50	3	6
E_R4_C-1_D2	DEAD	2	2012-07	2021-08	100	3	19
E_R4_C-4_D1	DEAD	1	2012-12	2021-08	47	2	20
W_R10_C4_D1	DEAD	1	2013-09	2021-08	50	3	21
W_R24_C1_D2	DEAD	2	2013-09	2021-08	62	3	21
W_R24_C1_D4	DEAD	4	2013-09	2021-08	62	4	21
E_R10_C1_D2	DEAD	2	2020-06	2021-09	4	4	22
W_R10_C-4_D1	DEAD	1	2013-09	2021-09	50	3	20
W_R4_C-4_D3	DEAD	3	2013-09	2021-09	50	4	20
C_R24_C-1_D1	DEAD	1	2013-09	2021-10	61	2	18
C_R4_C4_D3	DEAD	3	2013-09	2021-11	50	2	19
E_R10_C1_D1	DEAD	1	2012-07	2021-12	11	13	23
W_R18_C-1_D2	DEAD	2	2013-09	2021-12	66	3	19
C_R24_C1_D2	DEAD	2	2013-09	2022-02	65	4	18
C_R18_C1_D2	DEAD	2	2013-09	2022-03	66	2	16
E_R18_C-1_D3	DEAD	3	2012-07	2022-03	49	3	12
E_R18_C4_D2	DEAD	2	2012-07	2022-03	67	1	12
W_R4_C-1_D1	DEAD	1	2013-09	2022-03	50	3	17
C_R10_C1_D2	DEAD	2	2013-09	2022-04	50	2	13
C_R10_C-4_D1	DEAD	1	2013-09	2022-06	12	12	14
C_R10_C1_D1	DEAD	1	2013-09	2022-06	30	3	31
C_R18_C4_D2	DEAD	2	2013-09	2022-06	65	3	17
W_R24_C4_D3	DEAD	3	2013-09	2022-06	72	3	17
W_R4_C4_D1	DEAD	1	2013-09	2022-06	50	3	17
W_R24_C-1_D2	DEAD	2	2013-09	2022-07	73	3	17
W_R24_C-1_D4	DEAD	4	2013-09	2022-08	68	6	16
E_R24_C-1_D1	DEAD	1	2012-12	2022-11	62	1	10
E_R24_C4_D1	DEAD	1	2012-07	2022-12	67	1	10
C_R18_C-4_D4	DEAD	4	2013-09	2023-01	64	2	27
E_R24_C1_D1	DEAD	1	2012-07	2023-01	42	6	5
W_R18_C-4_D2	DEAD	2	2013-09	2023-01	69	7	14
C_R24_C4_D2	DEAD	2	2013-09	2023-02	62	4	20
C_R4_C-4_D3	DEAD	3	2013-09	2023-03	50	2	13
E_R24_C-4_D1	DEAD	1	2012-07	2023-03	67	1	9
E_R4_C-1_D4	DEAD	4	2012-07	2023-04	44	3	23
W_R4_C-4_D1	DEAD	1	2013-09	2023-06	35	5	13
C_R4_C1_D1	DEAD	1	2013-09	2023-11	50	2	11
E_R10_C4_D2	DEAD	2	2012-07	2023-11	102	1	14
W_R4_C4_D2	DEAD	2	2013-09	2023-11	50	3	11
W_R24_C-4_D2	DEAD	2	2013-09	2023-12	90	5	10
C_R10_C-4_D2	DEAD	2	2013-09	2024-03	50	2	9
E_R10_C-4_D1	DEAD	1	2012-07	2024-03	4	5	5
W_R24_C4_D1	DEAD	1	2013-09	2024-03	87	4	10
W_R24_C4_D2	DEAD	2	2013-09	2024-03	87	6	10
C_R4_C4_D1	DEAD	1	2013-09	2024-08	50	2	7
C_R18_C-1_D3	DEAD	3	2013-09	2024-11	62	5	7
C_R10_C-4_D4	DEAD	4	2013-09	2025-01	52	4	6
E_R10_C-1_D1	DEAD	1	2012-07	2025-01	151	0	5
C_R10_C4_D4	DEAD	4	2013-09	2025-04	55	2	5
C_R24_C-1_D4	DEAD	4	2013-09	2025-05	62	6	4
E_R24_C-1_D4	DEAD	4	2012-07	2025-05	87	3	3
E_R24_C1_D4	DEAD	4	2012-07	2025-05	87	1	3
C_R10_C-1_D2	INTERMITTENT	2	2013-09	2025-06	57	4	4

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Table 2 – continued

Sensor ID	Status	Depth	First	Last	Run	Res.	Plat. %
C_R18_C1_D1	INTERMITTENT	1	2013-09	2025-06	65	4	14
E_R10_C-4_D4	INTERMITTENT	4	2012-07	2025-06	62	8	5
W_R10_C-4_D2	INTERMITTENT	2	2013-09	2025-06	57	3	3
E_R10_C-1_D2	INTERMITTENT	2	2012-12	2025-07	103	1	6
C_R4_C1_D4	INTERMITTENT	4	2013-09	2025-08	43	6	8
E_R4_C1_D4	INTERMITTENT	4	2012-07	2025-08	62	2	15
W_R18_C4_D4	INTERMITTENT	4	2013-09	2025-08	110	3	2
W_R18_C1_D2	INTERMITTENT	2	2013-09	2025-09	111	3	4
E_R10_C4_D1	INTERMITTENT	1	2012-12	2025-11	156	0	2
E_R18_C1_D2	INTERMITTENT	2	2012-12	2025-11	72	2	5
C_R10_C-1_D3	INTERMITTENT	3	2013-09	2026-01	44	5	6
E_R24_C4_D3	INTERMITTENT	3	2012-07	2026-01	95	2	1
C_R4_C-4_D1	ALIVE_RECENT	1	2013-09	2026-03	66	3	4
W_R10_C-1_D2	ALIVE_RECENT	2	2013-09	2026-03	66	4	0
W_R10_C-4_D4	ALIVE_RECENT	4	2013-09	2026-03	50	4	10
W_R10_C1_D2	ALIVE_RECENT	2	2013-09	2026-03	66	3	0
W_R10_C1_D3	ALIVE_RECENT	3	2013-09	2026-03	50	4	11
W_R10_C4_D2	ALIVE_RECENT	2	2013-09	2026-03	50	4	15
W_R10_C4_D4	ALIVE_RECENT	4	2013-09	2026-03	66	3	0
W_R18_C-1_D3	ALIVE_RECENT	3	2013-09	2026-03	117	4	0
W_R18_C1_D1	ALIVE_RECENT	1	2013-09	2026-03	117	3	0
W_R4_C-1_D2	ALIVE_RECENT	2	2013-09	2026-03	50	4	7
W_R4_C1_D1	ALIVE_RECENT	1	2013-09	2026-03	51	4	2
W_R4_C1_D2	ALIVE_RECENT	2	2013-09	2026-03	66	10	0
W_R4_C1_D4	ALIVE_RECENT	4	2013-09	2026-03	50	4	10
C_R10_C-1_D1	ALIVE_RECENT	1	2013-09	2026-05	68	2	0
C_R10_C4_D1	ALIVE_RECENT	1	2013-09	2026-05	68	2	0
C_R10_C4_D2	ALIVE_RECENT	2	2013-09	2026-05	68	2	0
C_R18_C-1_D1	ALIVE_RECENT	1	2013-09	2026-05	62	4	0
C_R18_C-1_D2	ALIVE_RECENT	2	2013-09	2026-05	62	6	0
C_R18_C-4_D2	ALIVE_RECENT	2	2013-09	2026-05	62	4	0
C_R18_C1_D3	ALIVE_RECENT	3	2013-09	2026-05	65	4	1
C_R18_C4_D1	ALIVE_RECENT	1	2013-09	2026-05	62	4	0
C_R18_C4_D4	ALIVE_RECENT	4	2013-09	2026-05	65	3	0
C_R24_C-1_D2	ALIVE_RECENT	2	2013-09	2026-05	57	3	17
C_R24_C1_D4	ALIVE_RECENT	4	2013-09	2026-05	62	4	1
C_R4_C-1_D2	ALIVE_RECENT	2	2013-09	2026-05	68	6	6
C_R4_C-4_D2	ALIVE_RECENT	2	2013-09	2026-05	68	2	0
C_R4_C1_D2	ALIVE_RECENT	2	2013-09	2026-05	50	3	8
C_R4_C4_D2	ALIVE_RECENT	2	2013-12	2026-05	39	19	0
E_R10_C-4_D2	ALIVE_RECENT	2	2012-07	2026-05	114	2	0
E_R10_C1_D3	ALIVE_RECENT	3	2012-07	2026-05	112	2	7
E_R10_C4_D4	ALIVE_RECENT	4	2012-07	2026-05	113	3	0
E_R18_C-1_D1	ALIVE_RECENT	1	2012-07	2026-05	99	1	0
E_R18_C-1_D2	ALIVE_RECENT	2	2012-07	2026-05	99	3	0
E_R18_C-4_D2	ALIVE_RECENT	2	2012-07	2026-05	99	2	0
E_R18_C-4_D4	ALIVE_RECENT	4	2012-07	2026-05	99	2	0
E_R18_C1_D1	ALIVE_RECENT	1	2012-07	2026-05	42	6	11
E_R18_C1_D3	ALIVE_RECENT	3	2012-07	2026-05	99	3	0
E_R18_C4_D4	ALIVE_RECENT	4	2012-07	2026-05	99	1	0
E_R24_C-1_D2	ALIVE_RECENT	2	2012-07	2026-05	99	4	0
E_R24_C-4_D2	ALIVE_RECENT	2	2012-07	2026-05	99	1	0
E_R24_C-4_D3	ALIVE_RECENT	3	2012-07	2026-05	99	2	0
E_R24_C1_D2	ALIVE_RECENT	2	2012-07	2026-05	99	1	0
E_R4_C-1_D1	ALIVE_RECENT	1	2012-07	2026-05	135	1	6
E_R4_C-4_D3	ALIVE_RECENT	3	2012-07	2026-05	113	3	0
E_R4_C1_D1	ALIVE_RECENT	1	2012-07	2026-05	113	3	0
E_R4_C4_D1	ALIVE_RECENT	1	2012-07	2026-05	113	2	0
C_R10_C1_D3	stillborn	3	–	–	–	–	100
W_R10_C-1_D3	stillborn	3	–	–	–	–	100
W_R18_C-4_D4	stillborn	4	–	–	–	–	100

continued on next page

Table 2 – continued

Sensor ID	Status	Depth	First	Last	Run	Res.	Plat. %
W_R18_C4_D2	stillborn	2	–	–	–	–	100

4 Spatial Structure and the Channel Hypothesis

The depth pattern of the previous section invites a natural alternative explanation. Each LEO slope is built with a shallow central channel running downslope, a designed depression intended to concentrate water flow, and the soil surface model confirms it: a cross-section through the slope falls from roughly 3.38 m elevation at the outer columns to a thalweg near 2.77 m at the centre, a channel about 0.6 m deep and symmetric about the midline. If this channel concentrates moisture, and if moisture is what degrades the buried probes, then sensor failure should depend not only on depth but on cross-slope position, with the wet channel sensors failing preferentially. This is a confound that must be tested directly before the failure can be attributed to systemic hardware aging, because a spatial-moisture mechanism would mimic a common cause when compared across the three replicated slopes. The test is decisive and, on every axis examined, it does not support the hypothesis.

Taken first as a question about mortality, position carries no significant signal once depth is accounted for. The sensors nearest the channel ($|\text{Col}| = 1$) are dead in 50.0% of cases against 62.5% at the dry outer edge ($|\text{Col}| = 4$); the difference is not significant (Fisher $p = 0.18$) and, critically, it runs in the wrong direction for a wet-channel mechanism, with the drier edges failing slightly more, not less. A logistic regression that adds cross-slope distance and downslope row to a depth-only model improves the fit only marginally (likelihood-ratio $\chi^2 = 5.44$ on two degrees of freedom, $p = 0.066$), so spatial position does not add explanatory power beyond depth at any conventional threshold. The single large spatial contrast is the downslope one, where the topmost row is dead in 80.6% of cases against roughly 42% in the mid-slope rows, but this too points away from moisture: that row sits at the dry head of the slope, not the wet toe, and it is independently known to be an early instrumented trial row, so its extreme mortality is most parsimoniously an installation-cohort artefact rather than a hydrological signal. The most telling result is temporal: sensors near the channel and at the edge die at statistically indistinguishable times (median 2021-12 versus 2021-11, Mann–Whitney $p = 0.94$), so whatever kills them arrives as one slope-wide wave rather than propagating from the wet centre outward. Figure 8 collects this evidence.

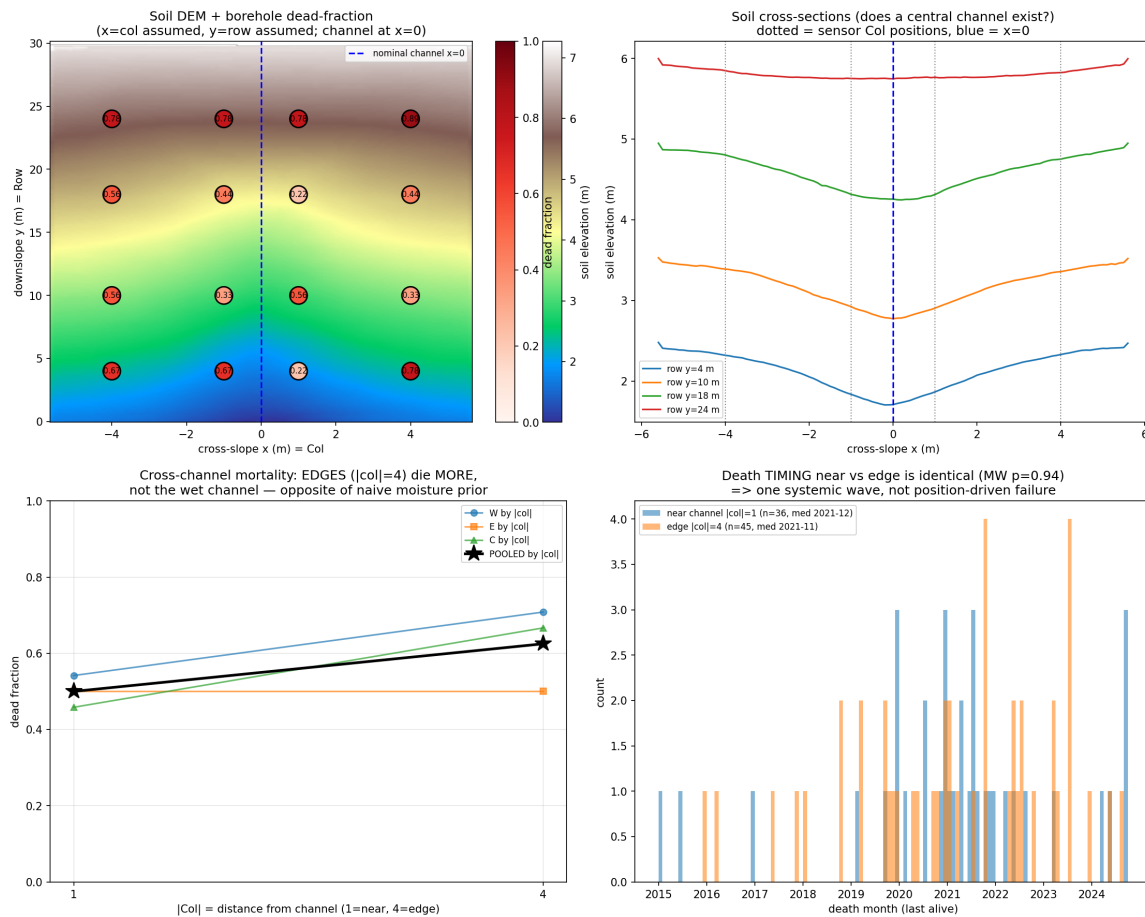


Figure 8: Spatial mortality against the soil surface model. The central channel is real (upper right, a 0.6 m symmetric depression), but mortality does not track proximity to it: the dry edges fail marginally more, not the wet channel (lower left), and death timing near the channel and at the edge is identical (lower right), the signature of one systemic wave rather than position-driven failure.

Because mortality is a coarse, low-count readout, the same hypothesis was put to the live signal, where it has its best chance and the statistical power is two orders of magnitude greater. Over the twelve-month window of peak array health, June 2014 to May 2015, the diurnal behaviour of 132 simultaneously living sensors was characterised by mean concentration, diurnal amplitude, and phase, and tested against cross-slope position with depth and slope held fixed. The channel leaves no detectable imprint. Mean carbon dioxide is 332.3 ppm near the channel against 335.7 ppm at the edge, a difference that vanishes in regression once depth is controlled ($p = 0.88$); diurnal amplitude is likewise flat at 22.3 against 22.6 ppm ($p = 0.64$); and concentration shows no relationship with the topographic height-above-thalweg wetness proxy ($r = 0.11$, $p = 0.19$). A direct check of the slope-centre against its perimeter, depth by depth, finds no significant gap at any level, including the shallowest, where an earlier and cruder analysis had reported a large centre-high contrast that the grounded re-derivation does not reproduce. Depth, by contrast, orders the concentrations cleanly, from 315 ppm at the shallowest level to roughly 351 ppm deeper down. Figure 9 shows the live-signal result.

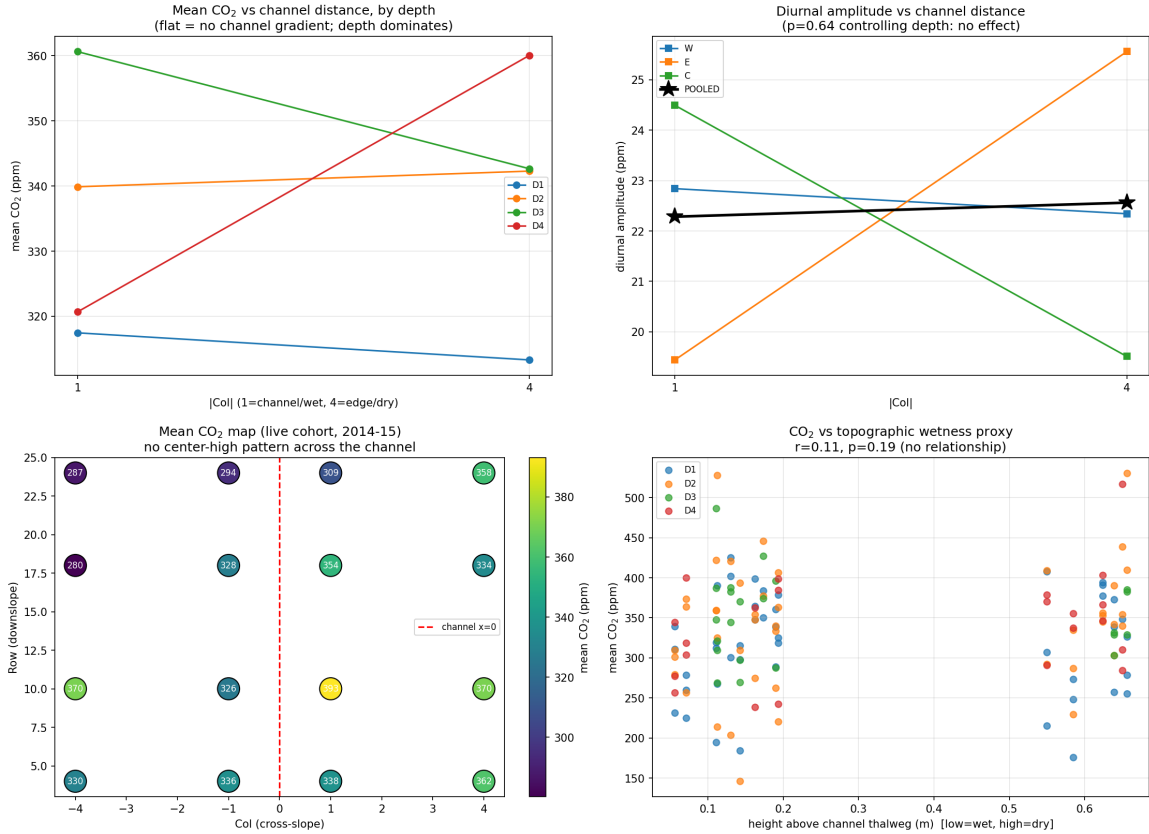


Figure 9: Live-signal test of the channel hypothesis on 132 sensors, 2014–2015. Mean concentration and diurnal amplitude are flat across the channel distance once depth is separated (top row), the spatial map shows no centre-high pattern (lower left), and concentration is uncorrelated with the topographic wetness proxy (lower right).

The channel is therefore a genuine topographic feature with no measurable expression in either sensor mortality or the live carbon-dioxide field once depth is separated. This is an informative null. It distinguishes the vertical structure of failure, which is real and surface-concentrated, from a lateral moisture structure, which is absent at the resolution these sensors provide, and it removes the most plausible spatial confound to the central conclusion: the dominant control on whether a probe survives is depth and the shared aging of the instrument population, not where the probe sits relative to the channel.

5 Spectral and Fourier Evidence

Gas transport through the basalt column is, to leading order, a linear diffusion process: the soil acts as a diffusive low-pass filter for the subsurface carbon-dioxide field, in direct analogy to thermal diffusion. Such a process leaves three independent fingerprints in the frequency domain that any genuine subsurface CO₂ channel must reproduce simultaneously. The first is the shape of the power spectral density (PSD): forced from the surface, the column behaves as a red-noise low-pass filter, so its spectrum follows a power law $S(f) \propto f^\beta$ over the diffusive band, with the diurnal forcing and its harmonics superimposed as a comb of sharp lines. The second is depth-dependent phase latency: because diffusion delays and attenuates the forced diurnal CO₂ wave as it propagates downward, each deeper sensor lags the shallower ones by a frequency-dependent phase, the deeper the larger. The third is inter-depth coherence: sensors in the same borehole are driven by one common surface forcing, so at the forcing frequencies their signals are nearly phase-locked. A reading can drift, saturate, or sit at an implausible value and still be defended as “unusual but real”; it cannot violate all three invariants at once and still be measuring the diffusion field. Spectra here are estimated by

Welch's method on 30-day Hann-windowed segments at a sampling rate of 96 samples per day, with the slope band taken over 5 h–20 d (0.05–4.8 cycles per day).

The contrast between a live and a dead channel is unambiguous (Figure 10). Over the window 2023-01-01 to 2026-05-16 the alive sensor E_R4_C1_D1 carries a clean power law with $\beta = -2.31$ ($R^2 = 0.90$ over 143 spectral points), while the dead sensor E_R4_C-4_D1 in the same row shows a flat, collapsed spectrum. At the diurnal line the alive PSD is 5434 against 169 for the dead channel, a factor of 32 in power.

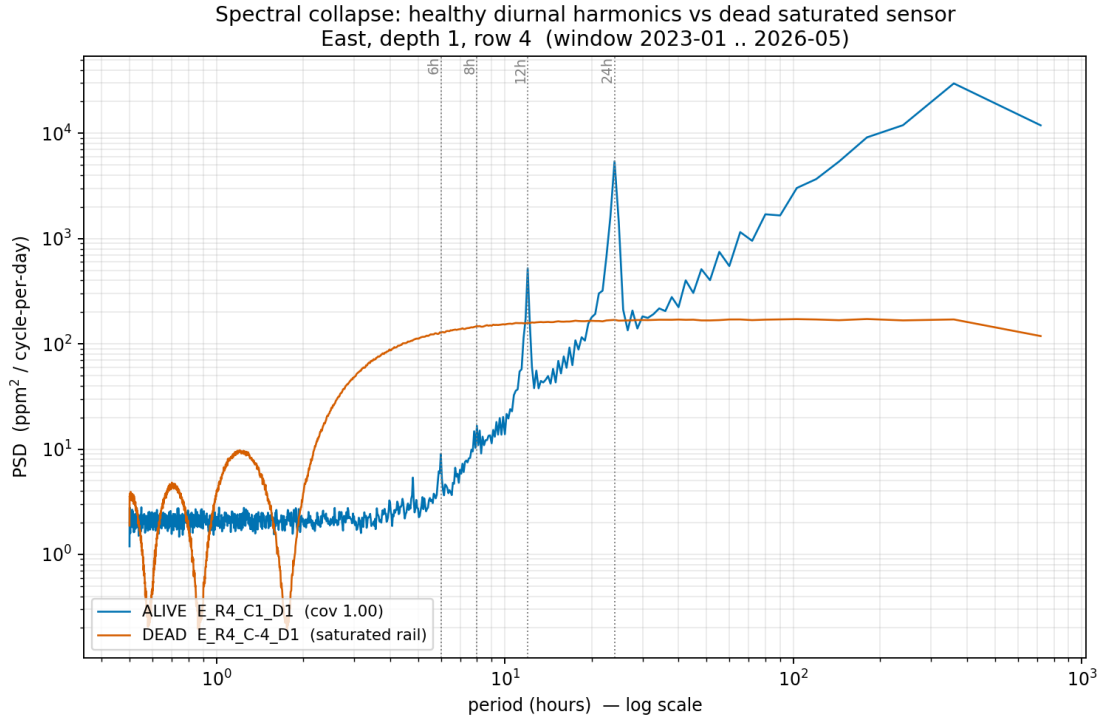


Figure 10: Alive versus dead power spectral density. The live channel follows the red-noise power law of diffusion; the dead channel is flat across all frequencies.

The diurnal comb of a healthy channel resolves into a structured set of lines (Figure 11). For E_R4_C1_D1 the fundamental at 24 h carries PSD 5434, the 12 h harmonic 523, the 8 h harmonic 16.6, and the 6 h harmonic 9.0, a monotone decay characteristic of a band-limited diffusive response. Superimposed on this natural comb are operational lines that are not of geophysical origin: a 3-day peak at PSD 952 consistent with the irrigation cadence, and a dominant 7-day peak at PSD 9153, the strongest single line in the spectrum, consistent with a weekly operational cycle. These operational attributions are inferred from the periods alone, no facility schedule having been cross-referenced here, and neither line bears on the liveness argument.

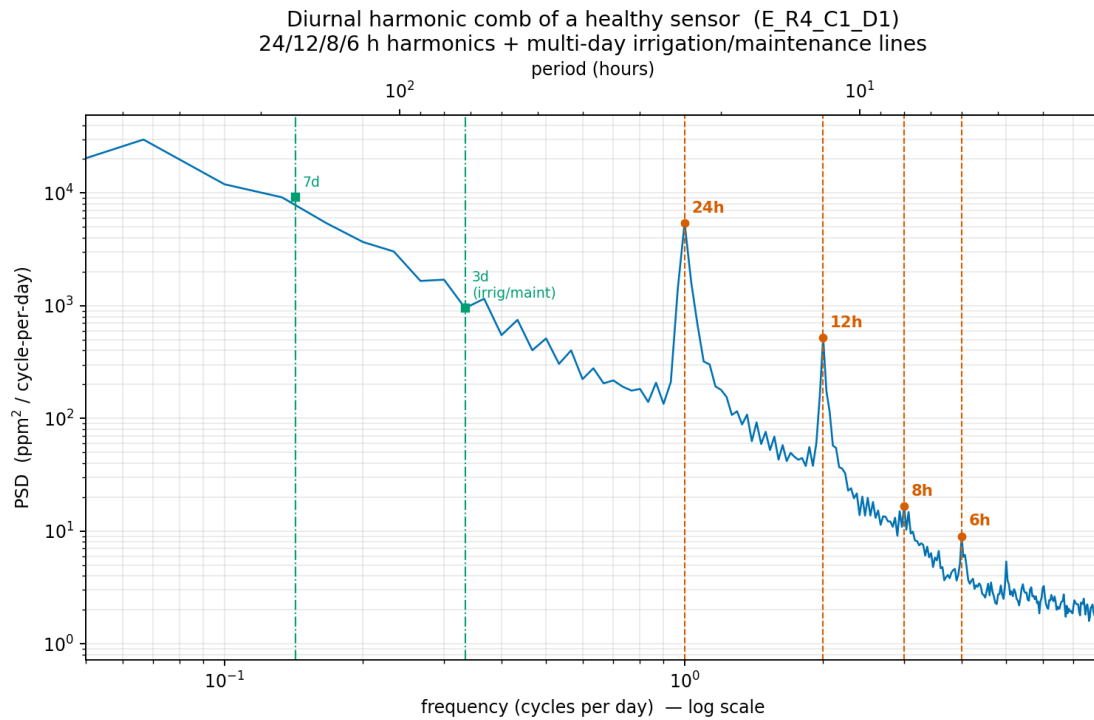


Figure 11: Diurnal harmonic comb of a healthy channel: the 24/12/8/6 h natural harmonics together with the 3-day irrigation and 7-day maintenance operational peaks.

The decisive evidence is what happens at the moment of failure (Figure 12). For W_R18_C4_D1, last alive on 2019-11-01, the year before failure has 99.99% plausible coverage and zero saturation, while the year after has only 5.07% plausible coverage and spends 94.9% of the year pinned to the saturation rail. Across the transition the diurnal 24 h peak collapses from 3528 to 21, a factor of 168, and the broadband signal power over the 5 h–20 d band drops by a median factor of 24. The collapse is essentially frequency-independent: both the diurnal line and the broadband floor fall together, so the entire spectral fingerprint flattens at once rather than losing a single band. These figures are computed on the cleaned, plausible-only channel; on the raw channel the post-failure output variance in fact rises, as the cell flaps between the saturation rail and brief excursions, which is itself a failure signature rather than any recovery of signal.

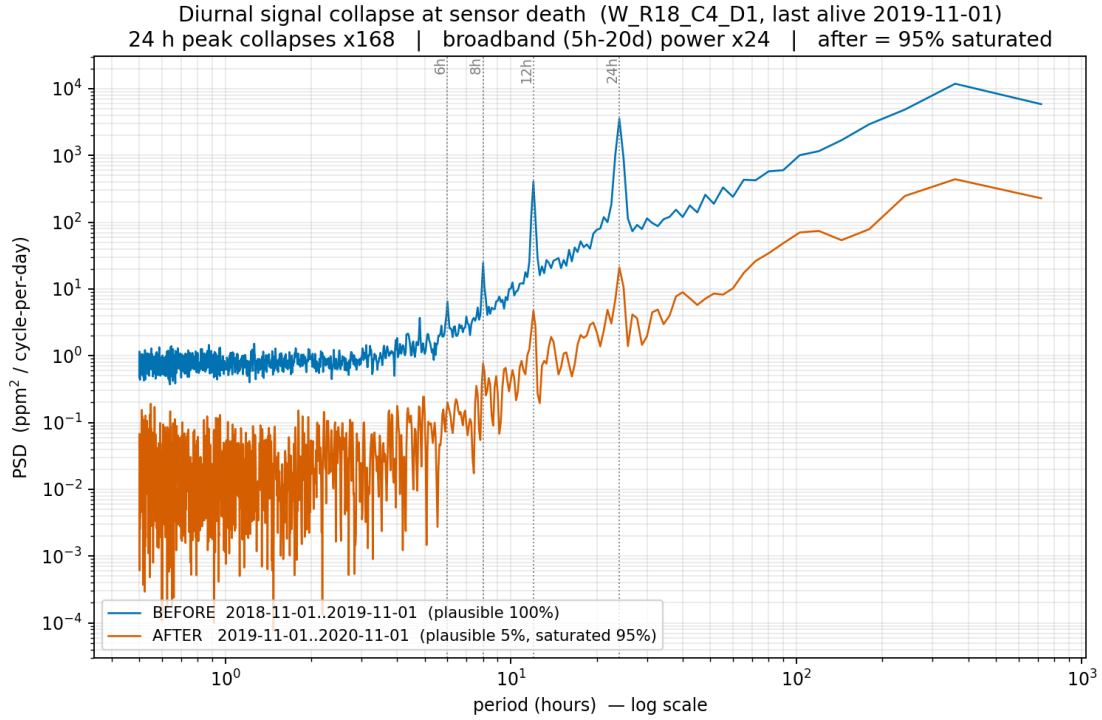


Figure 12: Before/after-failure spectra for a single channel. The 24 h peak collapses by a factor of 168 and broadband power by a factor of 24, a frequency-independent power collapse.

The third invariant fails in the same way (Figure 13). Two live sensors in one borehole, E_R18_C-1_D1 and E_R18_C-1_D2, are coherent at the diurnal frequency with magnitude-squared coherence 0.977. Pairing the same live sensor with the dead channel E_R18_C-1_D3 drops the diurnal coherence to 0.403: the dead channel no longer shares the common forcing of its own borehole. Healthy basalt thus obeys a power law with β between -2.31 and -1.64 (mean -1.98), measured on one representative healthy sensor per depth drawn from several boreholes, with inter-depth phase-locking demonstrated within a single borehole (E_R18_C-1); a dead sensor loses the power-law roll-off, the harmonic comb, and the inter-depth coherence together, and the simultaneity of these three failures is decisive evidence that it has stopped measuring the process.

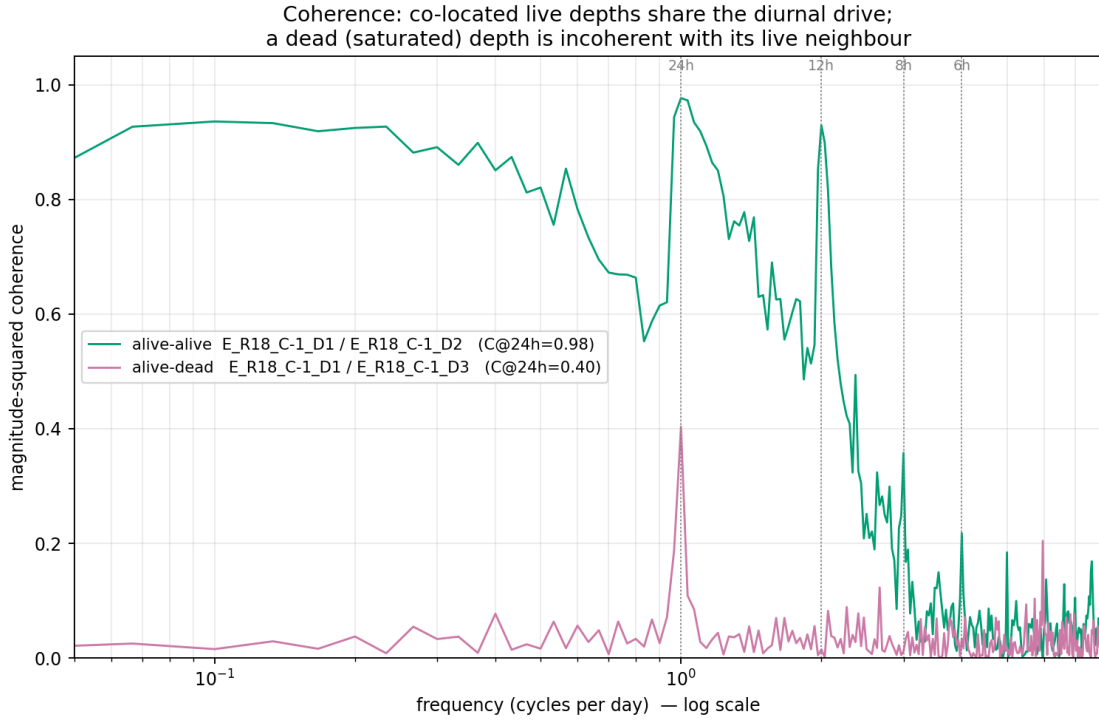


Figure 13: Inter-depth coherence. Two live channels are phase-locked at the diurnal frequency (0.977); a live–dead pair collapses to 0.403.

A per-channel spectral verdict for the whole array

The contrasts above are demonstrated on representative channels; to make the spectral argument carry the same weight as the liveness classification, the diurnal spectral diagnostic was computed for every one of the 144 channels rather than for examples alone. For each channel the diurnal spectral signal-to-noise ratio, defined as the power spectral density at the 24-hour line divided by the instrument noise floor in the 3 to 6-hour band, was measured over the most recent twelve months of record; a channel was judged spectrally alive when this ratio exceeds five, a level at which a genuine diurnal line stands clearly above the noise. The separation is clean. Recently-alive channels have a median diurnal ratio of 184, dead and stillborn channels a median of zero, and intermittent channels an intermediate 28. Of the 144 channels, 60 are spectrally alive, 76 show a collapsed diurnal line, and 8 are weak or never resolved.

The independent spectral verdict agrees with the time-domain liveness classification on 94% of the channels for which a single expectation applies (Table 3 and Figure 14). Every one of the 46 alive-recent channels is confirmed spectrally alive, and every one of the 4 stillborn channels is confirmed to carry no diurnal line. The eight disagreements are instructive rather than contradictory: all are channels the amplitude rule marks dead yet which retain a diurnal spectral line in the recent record, five of them strongly so, with channels such as C_R10_C1_D2, W_R4_C-1_D1 and E_R24_C1_D1 showing diurnal ratios from 211 to 854. These are channels whose diurnal cycle survives in the frequency domain but whose median amplitude or concentration falls outside the conservative liveness gate, so the amplitude rule errs on the side of declaring them dead. They are flagged here as the small set most deserving of a second look rather than treated as failures of either method. The full per-channel spectral table is provided in the accompanying `derived/spectral_classification_144.csv`.

Table 3: Amplitude-liveness status against the independent per-channel spectral verdict, all 144 channels. Agreement is 94% on the classes with a single expectation; the off-diagonal dead-yet-spectrally-alive channels are the ones the conservative amplitude gate under-counts.

Amplitude status	Spectrally alive	Spectrally collapsed	Weak / never
Alive-recent	46	0	0
Intermittent	9	3	1
Dead	5	73	3
Stillborn	0	0	4

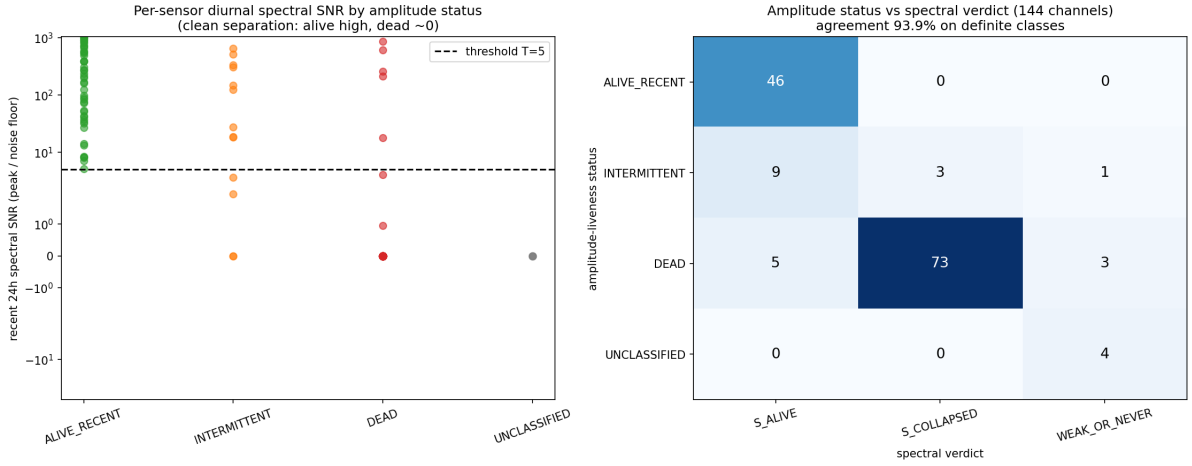


Figure 14: Per-channel diurnal spectral verdict for all 144 channels. Left: the recent diurnal spectral ratio by amplitude status, with the threshold at five; alive-recent channels sit far above it and dead channels at zero, the few exceptions being the dead-by-amplitude channels that retain a diurnal line. Right: the agreement matrix between the amplitude-liveness status and the spectral verdict.

6 Cross-Slope Synthesis

The three instrumented slopes were built as engineered replicas: identical construction, identical sensor type, and the same forty-eight basalt boreholes apiece. That symmetry is what makes them a controlled experiment. If failure were driven by something local to one slope — a drainage fault, a geochemical peculiarity, a single bad installation crew — it would show up as one slope dying differently from the other two. If instead failure is a property of the sensor type itself, it should reproduce across all three regardless of where the probe was buried. Comparing the slopes therefore distinguishes a systemic hardware vulnerability from a slope-local defect, and the comparison comes down decisively on the side of the former.

The magnitude of the failure is essentially the same everywhere. Of forty-eight basalt sensors per slope, West has lost thirty, Center twenty-seven, and East twenty-four; the distributions of when those deaths occurred are statistically indistinguishable between every slope pair (Mann–Whitney $p = 0.261$ for West–East, 0.137 for West–Center, 0.763 for East–Center), and a Kruskal–Wallis test across all three returns $H = 2.50$, $p = 0.286$. The cumulative failure curves never separate by more than 0.21 over the whole record. There is no slope that fails by a measurably different process or to a markedly different degree. The indistinguishable timing distributions, resting on only twenty-four to thirty deaths per slope, do not by themselves prove sameness, but read together with the strong year-by-year cross-slope correlation established below they are most consistent with a single shared aging process acting on the GMM222 probe population. Figure 15 shows the three trajectories overlaid.

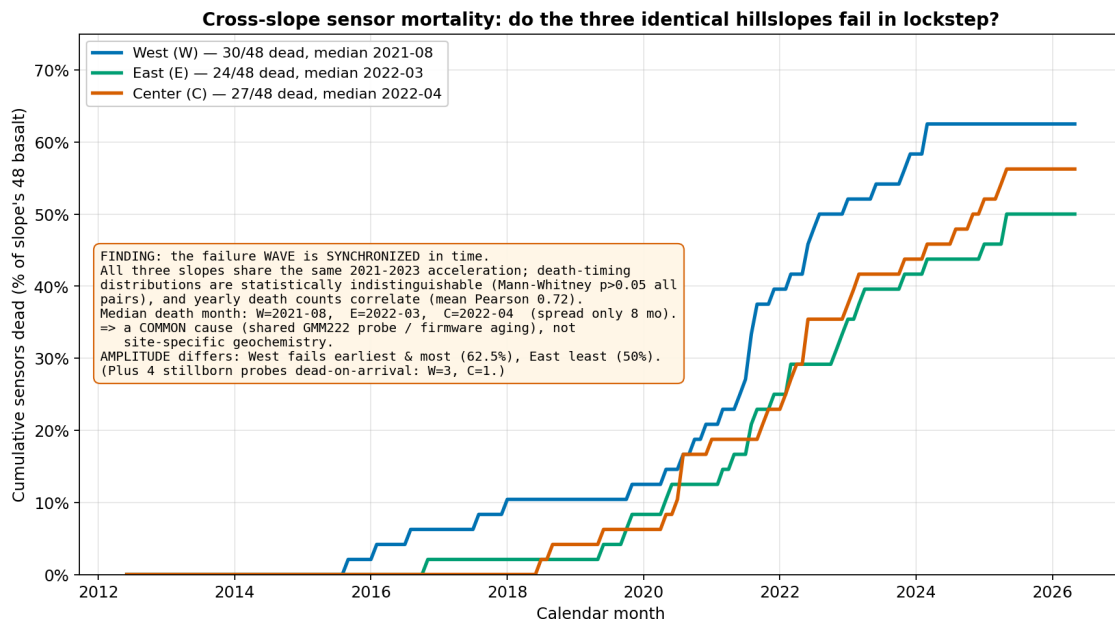


Figure 15: Failure across the three replica slopes. The magnitude and shape of the loss are common to all three; the individual death events are not time-locked.

What is common is the shape, not the schedule. The slopes do not fail in lockstep at the level of individual sensors. Month-to-month, the death counts are only weakly correlated across pairs (Spearman 0.362 for West–East, 0.189 for West–Center, 0.319 for East–Center), even though the same series viewed year-by-year track each other strongly (mean Spearman 0.739, mean Pearson 0.724). The reconciliation is straightforward: a common aging mechanism sets the multi-year trajectory each slope must follow, but each probe crosses its own failure threshold on its own clock. The median final-active month drifts from August 2021 on West to March 2022 on East and April 2022 on Center, a spread of eight months. This is the fingerprint of independent per-sensor firing under a shared cause, not of a synchronized external shock. The reading is reassuring in one respect and sobering in another: there is no looming cliff on which all remaining sensors fail together, but the degradation is intrinsic and will continue to claim sensors one by one.

East is the healthiest slope by every available measure. It retains eighteen of forty-eight sensors as recently active, the most of any slope; it has the fewest dead at twenty-four; and it is the only slope with no stillborn sensors — probes that never produced valid data — against three on West and one on Center. East’s recently-active fraction is 0.375 versus 0.333 for Center and 0.250 for West. The ordering matters because it is modest: a healthiest slope that is only marginally healthier than the worst is further evidence that the dominant cause is shared rather than local.

The depth-mortality pattern replicates only in part, and the part that replicates is the most important one. The shallowest horizon, depth D1, is the highest- or tied-highest-mortality depth on all three slopes without exception, which is what a moisture-driven mechanism concentrated near the surface predicts. Beyond that, the full depth gradient does not reproduce cleanly. West shows a steep shallow-over-deep decline with a pronounced D1 outlier (14 of 16 dead), East a near-monotonic fall with depth, and Center a flat-high profile across D1, D3, and D4 with D2 as the lone survivor horizon. The pairwise depth-profile correlations bear this out: West and East agree well (Pearson 0.764), but Center tracks neither (West–Center 0.293, East–Center -0.174). Figure 16 presents the three depth profiles side by side. The honest reading is partial replication: the surface-vulnerability signal is universal, while the finer depth structure is shaped by per-slope variation that the present data do not let us resolve further.

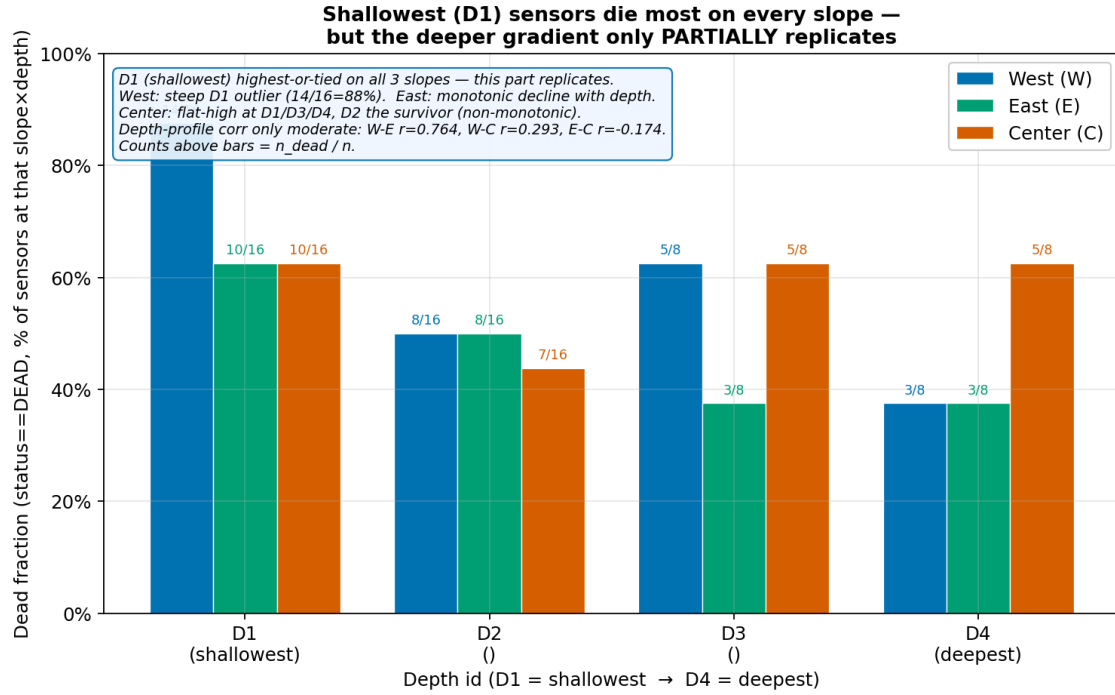


Figure 16: Depth-resolved mortality by slope. The shallowest horizon D1 is the worst-affected on every slope; the deeper gradient varies between slopes.

7 Recoverable Diffusion Signal (Forward Outlook)

This section is presented explicitly as exploratory and forward-looking, not as a headline conclusion of the audit. It is included to record what physical structure the surviving data can still support, and under what constraints, so that any future instrumentation programme begins from an honest baseline rather than from optimism. The governing constraint is stated first and without softening: at the end of the dataset zero boreholes retain a full three-depth profile of simultaneously live sensors, so the vertical diffusion structure cannot be probed on a currently-live column at all. It can only be reconstructed on historical windows during which a complete profile happened to remain alive. The analysis below therefore characterises a survivor, not an operational asset.

The window chosen for this probe is borehole W_R4_C-4 on the West slope, over the twelve months from 2017-10-01 to 2018-10-01, using the three install depths D1, D2 and D3. This borehole sits inside a longer interval, 2017-10 to 2020-08 (35 months), during which all three depths were simultaneously alive, which is what makes it usable. After cleaning to the plausible band, resampling to an hourly grid and despiking, roughly 3.8% of hourly bins were gap-interpolated per depth. On this reconstructed column three transport signatures survive, and they are mutually consistent in the direction physics requires.

The first signature is a monotonic 24-hour phase lag that increases with depth. The diurnal phasor peaks at local hour 14.34 at D1, 18.21 at D2 and 21.43 at D3, giving inter-depth latencies of 3.87 hours from D1 to D2, 3.22 hours from D2 to D3, and 7.09 hours from D1 to D3. The deeper sensor lags the shallower one in every pair, with no reversal. The second signature is amplitude attenuation with depth: the diurnal amplitude falls from 18.07 ppm at D1 to 17.62 ppm at D2 and 8.40 ppm at D3, so that D2 retains 0.9746 of the D1 amplitude and D3 only 0.4645. The third signature is the broadband spectral slope. Fitting the log-log power spectral density over the 5 to 480 hour band with the diurnal line excluded yields $\beta = -1.74$ at D1, -2.13 at D2 and -2.07 at D3, with fit R^2 of 0.95, 0.98 and 0.98 respectively. The two deeper channels sit close to the $\beta \approx -2$ expected of a diffusively smoothed background, and the shallower channel is somewhat flatter, consistent with a depth at which forcing has been filtered less. As an internal consistency check, regressing against ordinal depth index gives a log-amplitude slope of -0.383 per index and a

phase slope of 0.928 radians per index; for an ideal diffusion wave both would share one damping depth, and these are reported as comparable in direction rather than as a fitted diffusivity.

The caveats are stated plainly and they bound the claim hard. This borehole is a survivor, so the signal carries survivorship bias toward drier, longer-lived sensors and is not representative of a currently-live profile. The depth identifiers D1, D2 and D3 are an ordinal install index, not a verified absolute-centimetre ladder; inter-depth spacing is unknown, so neither latency-per-centimetre nor an absolute diffusivity can be derived. A single annual cycle precludes inter-year validation, so seasonal amplitude modulation is not averaged out and the diurnal phasor is a twelve-month mean. For these reasons no absolute diffusion coefficient is claimed here; this section reports transport signatures only, namely phase lag, amplitude attenuation and spectral slope.

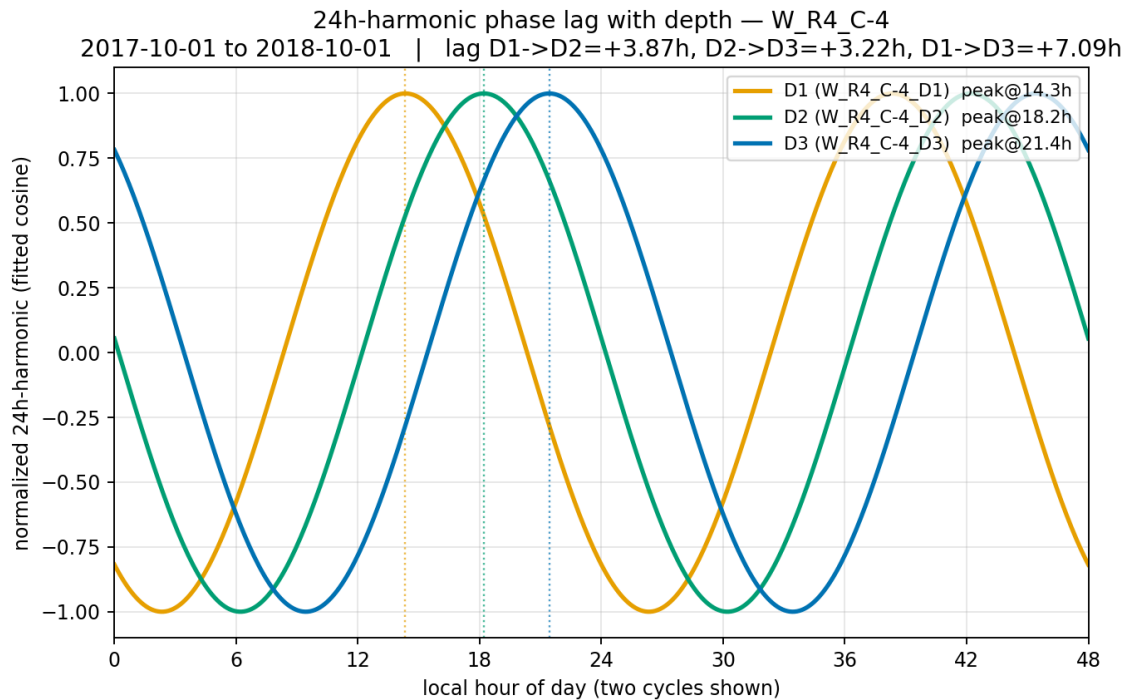


Figure 17: Monotonic 24-hour phase lag with depth on W_R4_C-4: the diurnal peak shifts from local hour 14.34 at D1 to 21.43 at D3, an accumulated lag of 7.09 hours over the profile.

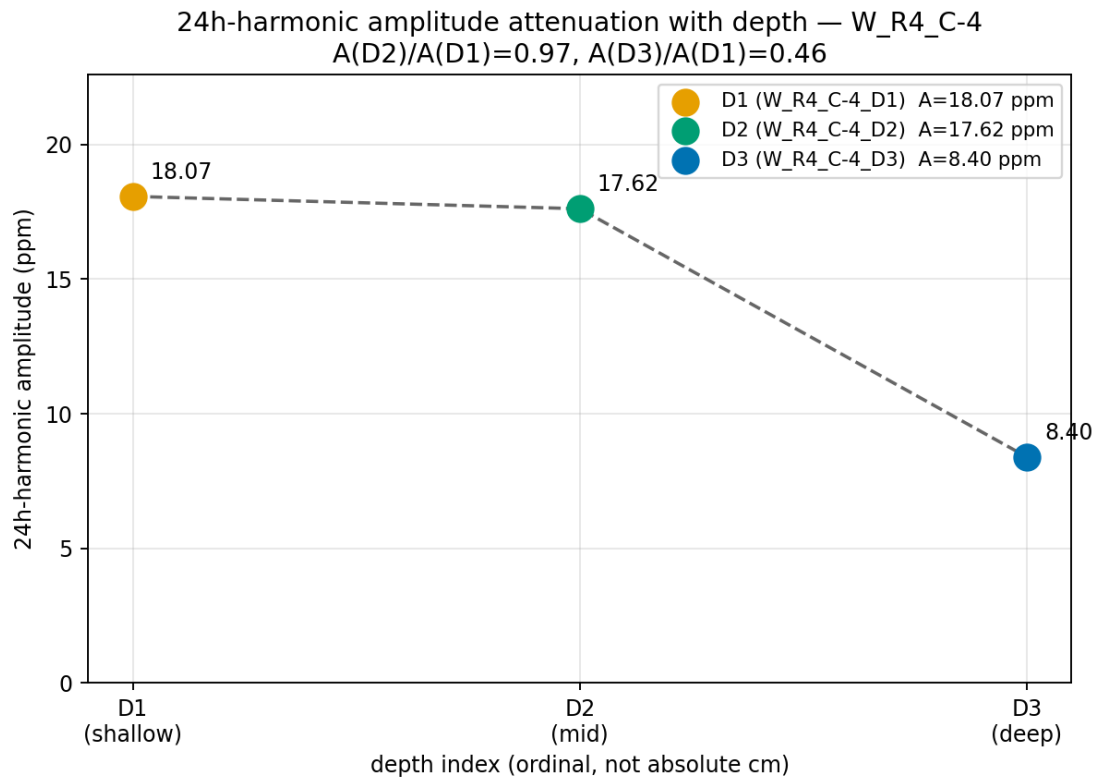


Figure 18: Diurnal amplitude attenuation with depth: 18.07 ppm at D1, 17.62 ppm at D2 and 8.40 ppm at D3, leaving 0.4645 of the surface amplitude at D3.

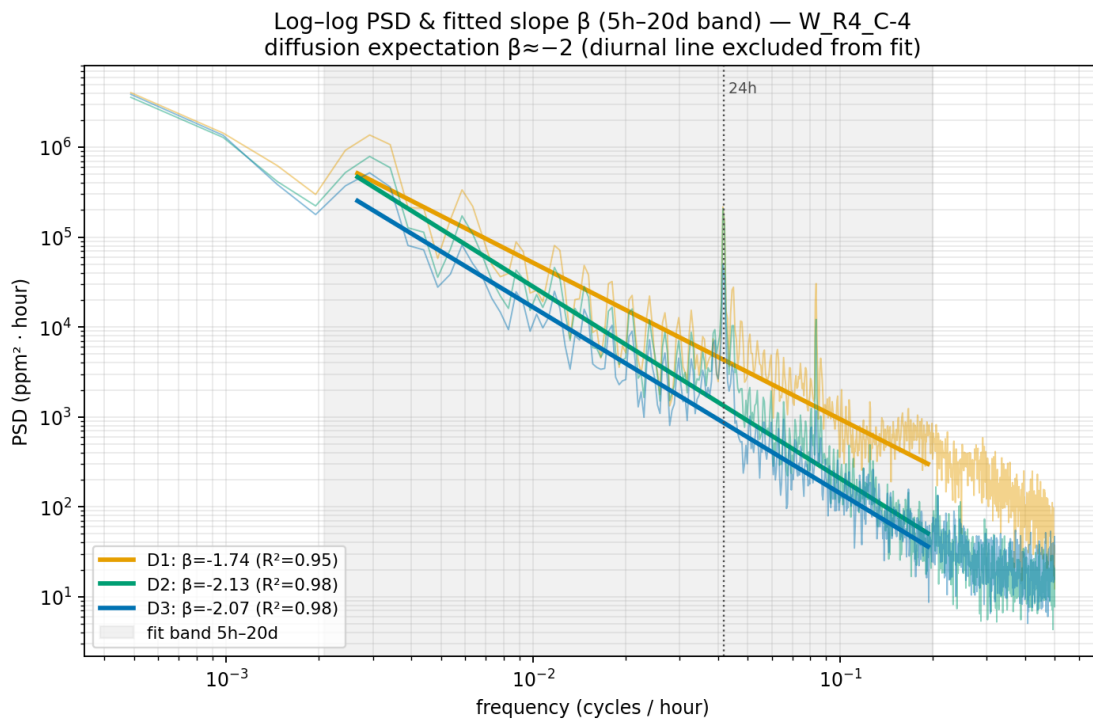


Figure 19: Log-log power spectral density slope over the 5–480 hour band (diurnal line excluded): $\beta = -1.74$, -2.13 and -2.07 at D1, D2 and D3, with the two deeper channels near the diffusive $\beta \approx -2$.

8 Conclusions and Limits

The basalt array as instrumented comprises 144 sensors across three slopes (48 each), with a record spanning roughly 13.8 years from June 2012 to May 2026. Its present state is not the binary working-or-broken picture an operator might assume from a raw data feed. Of the 144 channels, 81 are dead, 46 are alive in the recent record, 13 are intermittent, and 4 never produced a usable signal at all. Survival is uneven across slopes: East is the least degraded, retaining 18 of 48 alive and carrying the fewest dead at 24, while West is the worst at 12 alive and 30 dead, with Center between them at 16 alive and 27 dead. This verified status split, not the nominal channel count, is the array’s true operating capacity, and any planning that treats the network as 144 live probes overstates the available evidence by roughly a factor of three.

The dominant failure mechanism is systemic rather than local. Deaths are concentrated in a single epoch with median final-active months of 2021-08 on West, 2022-03 on East, and 2022-04 on Center, a spread of only eight months across the three slopes. Death-timing distributions are statistically indistinguishable between every slope pair (Mann–Whitney $p = 0.26, 0.14, 0.76$; Kruskal–Wallis $H = 2.50, p = 0.29$), and yearly death counts correlate positively across slopes (mean Pearson 0.72, mean Spearman 0.74), with the cumulative failure curves never separating by more than 0.21. A failure wave that follows a common multi-year trajectory on three geochemically distinct slopes cannot be driven by site-specific subsurface chemistry; it points to shared aging of the GMM222 probe population in a common hardware design, plausibly moisture-driven. The depth pattern is consistent with this reading: the shallowest horizon D1 is the highest-mortality depth on all three slopes, with a dead fraction of 0.708 pooled, against 0.479, 0.542, and 0.458 at depths 2, 3, and 4 — a surface-concentrated pattern of the kind a moisture agent would produce, though the deeper gradient is not strictly monotonic. Onset, however, is staggered per channel rather than collective, with deaths spanning 2015-09 to 2025-05, so the wave is a population-level aging process, not a single dated event.

The forensic record also explains why naive analysis windows are unsafe. Resurrection behaviour is pervasive: 134 of 144 sensors (93.1%) return to plausible output after apparent death at least once, the worst channel C_R4_C4_D2 doing so 19 times, and every basalt sensor carries GMM222 sentinel codes. A window chosen on the assumption that a channel is continuously live can therefore silently splice together genuine signal, saturated plateau, and sentinel flicker. The spectral evidence quantifies the cost of getting this wrong: a healthy channel follows a red-noise power law with slope in $[-2.31, -1.64]$ and a clean diurnal harmonic comb, whereas at death the 24-hour peak collapses by a factor of 168 and broadband power by 24, and a live–dead pair retains only 0.40 coherence at the diurnal line against 0.98 for a live–live pair. This report’s calendar-anchored protocol, which fixes windows by date and validates liveness against the verified status and resurrection record before any channel enters an analysis, removes precisely this contamination.

What is scientifically recoverable is correspondingly bounded. Healthy historical full profiles do exist and carry real transport structure: the West borehole W_R4_C-4 over 2017-10 to 2018-10 shows monotonic phase lag and amplitude decay with depth and a spectral slope near -2 at all three depths. But no borehole retains all three depths simultaneously alive at the dataset end, against a peak of 41 in a single month of 2014, so every recoverable profile is a survivor, and the depth index is ordinal rather than metric, which forbids an absolute diffusivity. A downstream modelling effort can responsibly use the East recent record and the calendar-anchored historical full profiles for relative transport signatures and spectral characterisation; it cannot treat the array as a currently complete depth-resolved network, nor infer absolute diffusion constants, without overstating what the surviving instruments support.